Midterm 01

REAL VARIABLES I

Fall 2026

MATH 55003 Real Variables I

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Name: _____

This exam has 5 problems worth 100 points total.

Instructions

This exam has several problems to choose from. You must choose 4 problems and solve them.

Solve exactly 4 problems. Only 4 attempted solutions will be graded. Clearly indicate which problems you decided to solve. All problems have the same weight.

You have 120 minutes (2 hours) from the moment you download this PDF to solve the problems. You then have an extra 15 grace period to photograph/scan and upload your solutions to gradescope.

Solutions must be hand-written. The exam is closed book. No notes, or external help is allowed.

Do not open this booklet until you are told to do so.

This booklet contains 11 pages in total.

Problem 1

Let $(X, \mathcal{A}, \mu)$ be a measure space and let $f_n : X \to \mathbb{R}$, $n \in \mathbb{N}$, be integrable functions such that

$$\sum_{n=0}^{\infty} \int_X |f_n(x)| \, \mathrm{d}\mu(x) < +\infty.$$

Prove that the limit $\sum_{n=0}^{\infty} f_n(x)$ exists and is finite for μ-almost every $x \in X$, and that, setting it to 0 where it does not exist or is infinite, the function $g(x) \coloneqq \sum_{n=0}^{\infty} f_n(x)$ satisfies

$$\int_X |g(x)| \, \mathrm{d}\mu(x) < +\infty.$$

Make sure to clearly state which limit theorems you use in your reasoning.

You may use the following fact about series: if

$$\sum_{n=0}^{\infty} |a_n| < +\infty,$$

then the series $\sum_{n=0}^{\infty} a_n$ converges to a finite value.

Space for continued solution.

Problem 2

- Prove that any set $E \subset [0,1]$ that is countable is a Borel set. (This fact is used in the next part.)
- Let $f : [0,1] \to \mathbb{R}$ have countably many points of discontinuity. Prove that f is Borel measurable.

Space for continued solution.

Problem 3

Consider the space $[0,1]$ and let $\mathcal{L}^*$ be the Lebesgue outer measure on $[0,1]$ and let $\mathcal{L}$ be the Lebesgue measure on $[0,1]$.

- Prove that for any set (measurable or not) $E \subset [0,1]$ and any $\varepsilon > 0$ there exists an open set $A \supseteq E$ such that $\mathcal{L}(A) \leq \mathcal{L}^*(E) + \varepsilon$.
- Prove that for every Borel set $E \subset [0,1]$ it holds that

$$\mathcal{L}(E) = \sup\{\mathcal{L}(K) : K \text{ is closed, and } K \subset E\}.$$

Hint: You may freely use that $K \subseteq E$ if and only if $E^c \subseteq K^c$. A set K is closed if and only if K^c is open. For any $A \subset \mathbb{R}$ it holds that $\sup A = -\inf(-A)$, where $-A = \{-x : x \in A\}$.

Space for continued solution.

Problem 4

Let $E \subset [0,1]$ be the collection of points $x \in [0,1]$ such that for every $n \geq 1$ there exists a rational $p/q \in [0,1]$ with $q \leq n$ such that $|x - p/q| < n^{-3}$. Prove that

- E is a Borel set.
- $\mathcal{L}(E) = 0$, where $\mathcal{L}$ is the Lebesgue measure.

Hint: Consider using (and proving) that there are at most n^2 rationals in $[0,1]$ with denominator at most n.

Space for continued solution.

Problem 5

Let X be a space and $\mathcal{A}$ a σ-algebra on X.

- Define what it means that $\mu : \mathcal{A} \to [0, +\infty]$ is a *measure*.
- Let $\mu : \mathcal{A} \to [0, +\infty]$ be a set functional and suppose the following properties hold.
  - It holds that $\mu(\emptyset) = 0$ and $\mu(X) = 1$.
  - For any $A, A' \in \mathcal{A}$ with $A \cap A' = \emptyset$ it holds that
  $$\mu(A \cup A') = \mu(A) + \mu(A').$$
  - For any sequence of sets $(A_k \in \mathcal{A})_{k \in \mathbb{N}}$ it holds that
  $$\inf_n \mu\Big(\bigcap_{k=0}^{n} A_k \Big) = \mu\Big(\bigcap_{k=0}^{\infty} A_k \Big).$$

  Prove that μ is a measure.

If you use any statement about continuity of μ on increasing/decreasing sequences of sets implying σ-additivity, you must prove it.

Space for continued solution.